



Global burden of cancer in children and adolescents aged 0–19 years, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023

GBD 2023 Childhood Cancer Collaborators*

Summary

Background Information on childhood cancer burden is crucial for effective cancer policy planning. Unfortunately, observed paediatric cancer data are not available in every country, and previous global burden estimates have not discretely reported several common cancers of childhood. We aimed to inform efforts to address childhood cancer burden globally by analysing results from the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023, which now include nine additional cancer causes compared with previous GBD analyses.

Methods GBD 2023 data sources for cancer estimation included population-based cancer registries, vital registration systems, and verbal autopsies. For childhood cancers (defined as those occurring at ages 0–19 years), mortality was estimated using cancer-specific ensemble models and incidence was estimated using mortality estimates and modelled mortality-to-incidence ratios (MIRs). Years of life lost (YLLs) were estimated by multiplying age-specific cancer deaths by the standard life expectancy at the age of death. Prevalence was estimated using survival estimates modelled from MIRs and multiplied by sequelae-specific disability weights to estimate years lived with disability (YLDs). Disability-adjusted life-years (DALYs) were estimated as the sum of YLLs and YLDs. Estimates are presented globally and by geographical and resource groupings, and all estimates are presented with 95% uncertainty intervals (UIs).

Findings Globally, in 2023, there were an estimated 377 000 incident childhood cancer cases (95% UI 288 000–489 000), 144 000 deaths (131 000–162 000), and 11·7 million (10·7–13·2) DALYs due to childhood cancer. Deaths due to childhood cancer decreased by 27·0% (15·5–36·1) globally, from 197 000 (173 000–218 000) in 1990, but increased in the WHO African region by 55·6% (25·5–92·4), from 31 500 (24 900–38 500) to 49 000 (42 600–58 200) between 1990 and 2023. In 2023, age-standardised YLLs due to childhood cancer were inversely correlated with country-level Socio-demographic Index. Childhood cancer was the eighth-leading cause of childhood deaths and the ninth-leading cause of DALYs among all cancers in 2023. The percentage of DALYs due to uncategorised childhood cancers was reduced from 26·5% (26·5–26·5) in GBD 2017 to 10·5% (8·1–13·1) with the addition of the nine new cancer causes. Target cancers for the WHO Global Initiative for Childhood Cancer (GICC) comprised 47·3% (42·2–52·0) of global childhood cancer deaths in 2023.

Interpretation Global childhood cancer burden remains a substantial contributor to global childhood disease and cancer burden and is disproportionately weighted towards resource-limited settings. The estimation of additional cancer types relevant in childhood provides a step towards alignment with WHO GICC targets. Efforts to decrease global childhood cancer burden should focus on addressing the inequities in burden worldwide and support comprehensive improvements along the childhood cancer diagnosis and care continuum.

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Introduction

Children diagnosed with cancer in high-income countries (HICs) are estimated to have a 5-year overall survival that exceeds 80%.^{1,2} Unfortunately, these positive outcomes have not translated globally, with 5-year net survival estimated to be less than 20% for children who develop cancer in some low-income and middle-income countries (LMICs).³ Factors contributing to the inequity in outcomes include multilevel challenges such as access to timely diagnosis, resource-optimised treatments, and supportive care. In recognition of the major disparities and

preventable deaths associated with childhood cancer, WHO initiated the Global Initiative for Childhood Cancer (GICC)⁴ at the UN General Assembly in 2018, with the goal of improving childhood cancer outcomes and reducing suffering due to childhood cancer globally.

To effectively address childhood cancer globally, it is crucial to understand the burden of childhood cancers at the global, regional, national, and community levels. However, disease surveillance systems that record and report cancer burden data, such as population-based cancer registries and vital registration systems, are not

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Research in context

Evidence before this study

We searched PubMed for English-language articles estimating global childhood cancer burden and published between Jan 1, 2010, and Oct 15, 2023, using the terms “childhood or child or pediatric” and “cancer or malignancy or tumor or neoplasm or oncology” and “global or international or world or worldwide” and “burden or incidence or mortality or survival or prevalence or DALYs”. We identified separate studies that presented global estimates of select measures of childhood cancer burden, including incidence, mortality, and survival, each focused on estimates for one calendar year. Analyses of childhood cancer burden estimates from previous rounds of the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) have been published, which incorporate multiple measures including incidence, mortality, prevalence, years of life lost, years lived with disability, and disability-adjusted life-years (DALYs). Each of the aforementioned publications presented important aspects of childhood cancer burden, but no previous publication analysed global childhood cancer burden estimates inclusive of index cancers identified for monitoring progress of the WHO Global Initiative for Childhood Cancer (GICC) across multiple measures and time.

Added value of this study

To our knowledge, GBD 2021 was the first study to comprehensively and separately quantify the global burden of several cancer types which are relevant in childhood and adolescence, and our analysis of GBD 2023 results is the first formal GBD analysis of these additional cancer types in the child and adolescent age group. These cancer types include retinoblastoma, neuroblastoma, Burkitt lymphoma, soft tissue sarcomas, bone cancers, and hepatoblastoma, and their addition to GBD allows for comparisons to other cancer types and diseases estimated with cohesive methodology and across time. Estimates of these cancer types will contribute to a more

complete understanding of childhood cancer burden on global, regional, and national scales and are steps towards the ability to track progress towards WHO GICC targets globally. In addition, this is the first GBD analysis of changes in global childhood cancer burden over time, presenting estimates from 1990 to 2023. Our analysis suggests that the WHO GICC target cancer types comprised approximately half of childhood cancer deaths globally in 2023, with variation by world region. While the total number of childhood cancer deaths globally decreased between 1990 and 2023, it increased in the WHO African region, and premature loss of life due to childhood cancer was greatest in countries with the least resources in 2023. Childhood cancer was in the top ten causes of childhood death globally in 2023, and when compared with adult cancers, was the second-leading cause of cancer DALY burden in countries with a low Socio-demographic Index in 2023.

Implications of all the available evidence

Childhood cancers contribute substantial burden within global child health and global oncology. In countries with the least resources, children with cancer still suffer the greatest premature mortality, highlighting that while progress has been made in outcomes globally over the past several decades, disparities remain. The WHO GICC is a crucial collaborative capacity building and implementation strategy that has fundamentally changed the landscape of childhood cancer since it was established in 2018. The target cancers identified for monitoring of progress in the WHO GICC comprise a sizeable portion of childhood cancer burden globally, but a considerable portion of childhood cancer burden is not part of the identified targets, and this is greater in some world regions than others. These findings reflect the need for WHO GICC efforts to address strengthening of health systems and childhood cancer needs and priorities as a whole in order to implement equitable and sustainable improvements in childhood cancer outcomes.

available everywhere in the world, and there is a particular dearth of reliable data in LMICs.^{5,6} Previous studies have analysed the international burden of childhood cancer with a focus on incidence,⁷ incidence and mortality,⁸ and survival⁹ where data are available. Model-based estimates of the global burden of childhood cancer, which aim to provide estimates where data do not exist as well as estimates at a global level, have been produced for incidence,¹⁰ incidence and mortality,^{11,12} and survival.^{3,13}

The Global Burden of Diseases, Injuries, and Risk Factors Study (GBD)¹⁴ generates global estimates for diseases, including cancers, across time and incorporates improvements in each iteration, offering a sustainable framework for childhood cancer burden estimation. These estimates support crucial cancer control work worldwide while ongoing efforts to expand cancer surveillance systems continue.^{15,16} A previous analysis of GBD 2017

estimates of the global burden of childhood cancers presented results using multiple measures with a focus on disability-adjusted life-years (DALYs), a measure that incorporates both the morbidity and premature mortality of a disease.¹¹ However, with the advent of the WHO GICC a year later, monitoring gaps remained. In particular, incidence, mortality, and DALY estimates for only three of the six index GICC childhood cancers were available in the GBD 2017 analysis. Additionally, only approximately 73·5% of childhood cancer DALYs were categorised in the previous analysis, as important childhood cancer types were not separately reported in the GBD 2017 cause list.

In this analysis, we aimed to summarise the updated estimates of global childhood cancer burden in 2023 and across time from 1990 to 2023. These results present, for the first time in a formal GBD analysis of childhood and adolescent cancer burden, an additional nine new cancer causes, representing closer alignment with the six GICC

index cancers. In aggregate, these results add important health policy data to support health delivery considerations. This manuscript was produced as part of the GBD Collaborator Network and in accordance with the GBD Protocol.¹⁷

Methods

Overview

GBD provides a framework for estimating global disease burden and aims to produce comprehensive and comparable measures of disease burden across geography and time. GBD is updated at regular intervals, with each round incorporating new data, improving methods, and generating updated estimates for all included locations, age groups, sexes, and years. Updated GBD estimates thus supersede the estimates of previous rounds. We present estimates produced by the most recent round, GBD 2023,^{14,18,19} in this analysis. Details on GBD 2023 data sources are available on the Global Health Data Exchange (GHDx) website for GBD 2023. The study is compliant with the GATHER statement.²⁰ GBD used de-identified data, and the waiver of informed consent was reviewed and approved by the University of Washington Institutional Review Board (study number 9060).

Cancer estimation in GBD

Data sources for cancer estimation in GBD include population-based cancer registries, vital registration systems, and verbal autopsy data (appendix 1 pp 22–51; maps of site-years of cancer data available in GBD 2023 are available in appendix 2 pp 2–4).²¹ Cancer mortality estimates are generated using cancer-specific and sex-specific Cause of Death Ensemble models (CODEm).²² CODEm uses all available mortality data and cause-specific covariates to estimate mortality for each age group, sex, location, and year estimated in GBD 2023, selecting the optimal model based on out-of-sample predictive validity (appendix 1 pp 59–74). Mortality data included in cancer CODEm models include vital registration system and verbal autopsy data, as well as population-based cancer registry incidence data that have been transformed into mortality data through mortality-to-incidence ratios (MIRs). This transformation allowed for increased data availability, particularly for locations where mortality data were scarce. In GBD 2023, cancer registry data were adjusted to align with vital registration system data; where cancer registry data directly overlapped with vital registration data, cancer registry data were excluded from CODEm models.²¹ MIRs were primarily estimated using a spatiotemporal Gaussian process regression (ST-GPR).¹⁸ For hepatoblastoma and retinoblastoma, two cancers with lower data availability and relatively fewer deaths, MIRs were estimated using negative binomial models. ST-GPR modelling used incidence and mortality data from population-based cancer registries, while negative binomial models used cancer registry incidence and mortality data as well as

vital registration system mortality data to increase the data available for retinoblastoma (appendix 1 pp 56–58). Cancer cause-specific mortality estimates generated using CODEm were scaled to independently modelled all-cause mortality to produce final cancer mortality estimates. Years of life lost (YLLs) were estimated by multiplying the number of cancer-specific deaths in a given age group by the standard life expectancy at the age of death. To evaluate the relationship between age-standardised YLLs and Socio-demographic Index (SDI), we fitted a linear regression model and calculated the Pearson correlation.

Non-fatal cancer estimation begins by transforming final cancer mortality estimates to incidence estimates by dividing the mortality estimates by their corresponding MIRs.²¹ Prevalence across 10 years was modelled using incidence estimates and modelled survival based on MIRs, then divided into phases of cancer care to estimate years lived with disability (YLDs). Phases of cancer care differed for cohorts based on whether they were estimated to survive or not. For the cohort surviving for 10 years after their cancer diagnosis, two phases were estimated: (1) diagnosis and treatment and (2) remission. For the cohort not surviving for 10 years after their cancer diagnosis, four phases were estimated: (1) diagnosis and treatment, (2) remission, (3) metastatic or disseminated, and (4) terminal. The prevalence of each phase was multiplied by a phase-specific disability weight to generate YLD estimates. Disability weights are meant to represent the magnitude of health loss associated with health outcomes and range from 0 (full health) to 1 (equivalent to death; appendix 1 p 81). The estimates of YLDs and YLLs were summed to generate DALY estimates.

All point estimates in this analysis represent the mean of the estimated draws and are reported with corresponding 95% uncertainty intervals (UIs), estimated as the 2.5th and 97.5th percentiles of 250 draws at each cancer estimation step and propagated through each step (appendix 1 pp 51–81). More detailed method descriptions can be found in appendix 1 and in the broader GBD 2023 capstone publications.^{14,18,19,21}

Reporting definitions

Childhood cancers were defined as those occurring at ages 0–19 years for this analysis. As some countries and organisations define childhood cancers as occurring at ages 0–14 years, results are available for this age range and more specific age group subsets in the GBD Results Tool. Cancers as defined in ICD-10 chapter II,²³ C00–C97, are included in this analysis (details on cancer types estimated and their corresponding definitions are included in appendix 1 pp 12–21). Other malignant epithelial neoplasms were not a GBD cancer cause but were grouped together for the purposes of this analysis in an effort to present results in similar categories as the International Classification of Childhood Cancer, 3rd

For GBD 2023 data sources see
<https://ghdx.healthdata.org/gbd-2023/sources>

See Online for appendix 1

See Online for appendix 2

For the GBD Results Tool see
<https://vizhub.healthdata.org/gbd-results>

edition (ICCC-3),²⁴ and included the following GBD cancer causes: bladder cancer, breast cancer, cervical cancer, colon and rectum cancer, lip and oral cavity cancer, malignant skin melanoma, nasopharynx cancer, pancreatic cancer, stomach cancer, thyroid cancer, and tracheal, bronchus, and lung cancer. For results of cancer burden presented in alignment with the WHO GICC, the GBD cause brain and central nervous system (CNS) cancers was used in place of the WHO indicator cancer low-grade gliomas, and the GBD cause kidney cancer was used in place of the more specific WHO indicator cancer Wilms tumour. Results are presented by various groupings, including GBD world super-regions, SDI quintiles,¹⁸ WHO world regions,²⁵ and World Bank income groupings²⁶ (appendix 2 pp 8–12). Rates are reported per 100 000 person-years, with the GBD 2023 world standard population used for calculation of age-standardised rates (appendix 1 pp 81–82). GBD 2023 provides estimates for additional cancer types, measures, and groupings that are not included in this Article but are available online through the GBD Results Tool, GBD Compare, and GBD Cancer Compare visualisations.

Role of the funding source

The funders of the study had no role in study design, data collection, data analysis, data interpretation, or the writing of the report.

Results

There were 377 000 incident cases of childhood cancers (95% UI 288 000–489 000), 144 000 deaths (131 000–162 000), and 11·7 million (10·7–13·2) DALYs estimated due to cancer in children globally in 2023 (table). These DALYs were composed of a greater number of YLLs (11·5 million [10·5–13·0]) than YLDs (176 000 [108 000–279 000]; appendix 2 p 18). While higher SDI settings had the greatest age-standardised incidence rates for childhood cancer, they generally had the lowest age-standardised mortality rates (table). The majority of childhood cancer burden occurred in settings with limited resources, with 85·1% (82·9–87·5) of incident cases, 94·1% (93·5–94·8) of deaths, and 94·1% (93·4–94·7) of DALYs occurring in LMICs (comprising low-income, lower-middle-income, and upper-middle-income countries by World Bank grouping; appendix 2 p 15). The WHO Western Pacific region and African

For GBD Compare see <https://vizhub.healthdata.org/gbd-compare/>

For GBD Cancer Compare see <https://vizhub.healthdata.org/gbd-compare/cancer>

	Incident cases, thousands	Age-standardised incidence rate per 100 000 person-years	Percentage of global cases	Deaths, thousands	Age-standardised mortality rate per 100 000 person-years	Percentage of global deaths	DALYs, millions	Age-standardised DALY rate per 100 000 person-years	Percentage of global DALYs
Global	377 (288–489)	14·3 (10·9–18·6)	100·0% (100·0–100·0)	144 (131–162)	5·4 (4·9–6·1)	100·0% (100·0–100·0)	11·7 (10·7–13·2)	443·6 (404·0–498·0)	100·0% (100·0–100·0)
WHO regions									
African region	91·3 (74·6–112)	14·3 (11·7–17·6)	24·4% (21·2–27·8)	49·0 (42·6–58·2)	7·7 (6·7–9·1)	34·1% (31·8–36·6)	4·01 (3·50–4·77)	628·6 (546·9–747·0)	34·2% (31·8–36·9)
Eastern Mediterranean region	52·4 (40·1–68·0)	16·1 (12·3–20·9)	13·9% (12·7–14·8)	21·9 (18·5–25·0)	6·7 (5·7–7·7)	15·2% (13·9–16·3)	1·79 (1·51–2·04)	548·1 (463·0–626·1)	15·2% (13·9–16·3)
European region	40·0 (27·6–55·6)	18·6 (12·7–25·9)	10·6% (9·0–11·9)	8·02 (7·64–8·38)	3·7 (3·5–3·8)	5·6% (5·0–6·1)	0·660 (0·627–0·690)	303·6 (288·3–318·2)	5·6% (5·0–6·2)
Region of the Americas	49·5 (37·5–67·5)	17·1 (12·9–23·4)	13·1% (11·8–14·4)	14·3 (13·6–14·8)	4·9 (4·6–5·1)	10·0% (8·9–10·9)	1·16 (1·11–1·20)	397·9 (379·4–413·2)	9·9% (8·8–10·8)
South-East Asia region	49·2 (38·6–63·1)	7·9 (6·2–10·2)	13·1% (11·3–14·6)	23·3 (19·8–27·6)	3·7 (3·2–4·4)	16·2% (14·4–17·9)	1·88 (1·59–2·22)	303·7 (257·3–356·0)	16·0% (14·2–17·7)
Western Pacific region	93·0 (65·1–135)	17·3 (12·1–25·0)	24·5% (21·7–27·8)	26·9 (24·2–31·1)	4·8 (4·3–5·5)	18·7% (17·1–20·3)	2·19 (1·99–2·53)	396·7 (358·7–453·9)	18·7% (17·1–20·3)
SDI quintiles									
High	102 (69·0–149)	18·6 (12·5–27·4)	26·8% (23·4–30·6)	19·5 (18·3–20·9)	3·4 (3·2–3·7)	13·6% (12·3–14·7)	1·60 (1·50–1·71)	284·0 (266·1–305·8)	13·6% (12·3–14·9)
High-middle	73·6 (54·1–102)	16·4 (12·0–22·7)	19·5% (18·2–21·1)	23·7 (21·9–26·0)	5·1 (4·8–5·6)	16·5% (15·5–17·3)	1·92 (1·78–2·10)	421·4 (391·4–460·8)	16·4% (15·4–17·3)
Middle	46·3 (35·2–61·4)	14·0 (10·6–18·6)	12·3% (11·7–12·8)	18·5 (16·9–20·8)	5·6 (5·1–6·2)	12·9% (12·3–13·5)	1·51 (1·37–1·69)	453·8 (412·8–508·4)	12·9% (12·2–13·5)
Low-middle	47·2 (38·2–57·5)	10·4 (8·4–12·7)	12·6% (11·1–13·7)	23·1 (20·1–26·7)	5·1 (4·4–5·9)	16·1% (15·1–17·0)	1·87 (1·62–2·15)	414·3 (358·8–478·7)	16·0% (15·0–16·9)
Low	108 (90·3–133)	12·7 (10·6–15·6)	28·8% (24·9–32·3)	58·8 (51·7–68·7)	6·9 (6·1–8·1)	40·9% (39·0–43·3)	4·81 (4·22–5·61)	566·1 (496·8–660·9)	41·0% (39·0–43·4)

Values in parentheses are 95% uncertainty intervals. Estimates are presented to three significant figures for counts, or to one decimal place for rates and percentages. DALYs=disability-adjusted life-years. SDI=Socio-demographic Index.

Table: Global burden of childhood cancer in 2023, globally and by WHO region and SDI quintile, for all sexes combined

region had the most childhood cancer cases, while the WHO African region had the most childhood cancer deaths (table). Deaths due to childhood cancer decreased by 27·0% (15·5–36·1) globally between 1990 and 2023, from 197 000 (173 000–218 000) to 144 000 (131 000–162 000), but increased in the WHO African region by 55·6% (25·5–92·4), from 31 500 (24 900–38 500)

to 49 000 (42 600–58 200; table; appendix 2 p 32; GBD Results Tool). From 1990 to 2023, overall childhood deaths decreased by 56·5% (55·8 to 57·3) globally and childhood cancer incident cases changed by –0·9% (–27·1 to 38·6; GBD Results Tool; appendix 2 p 26). Age-standardised mortality rates decreased between 1990 and 2023 globally and in all WHO world regions, most

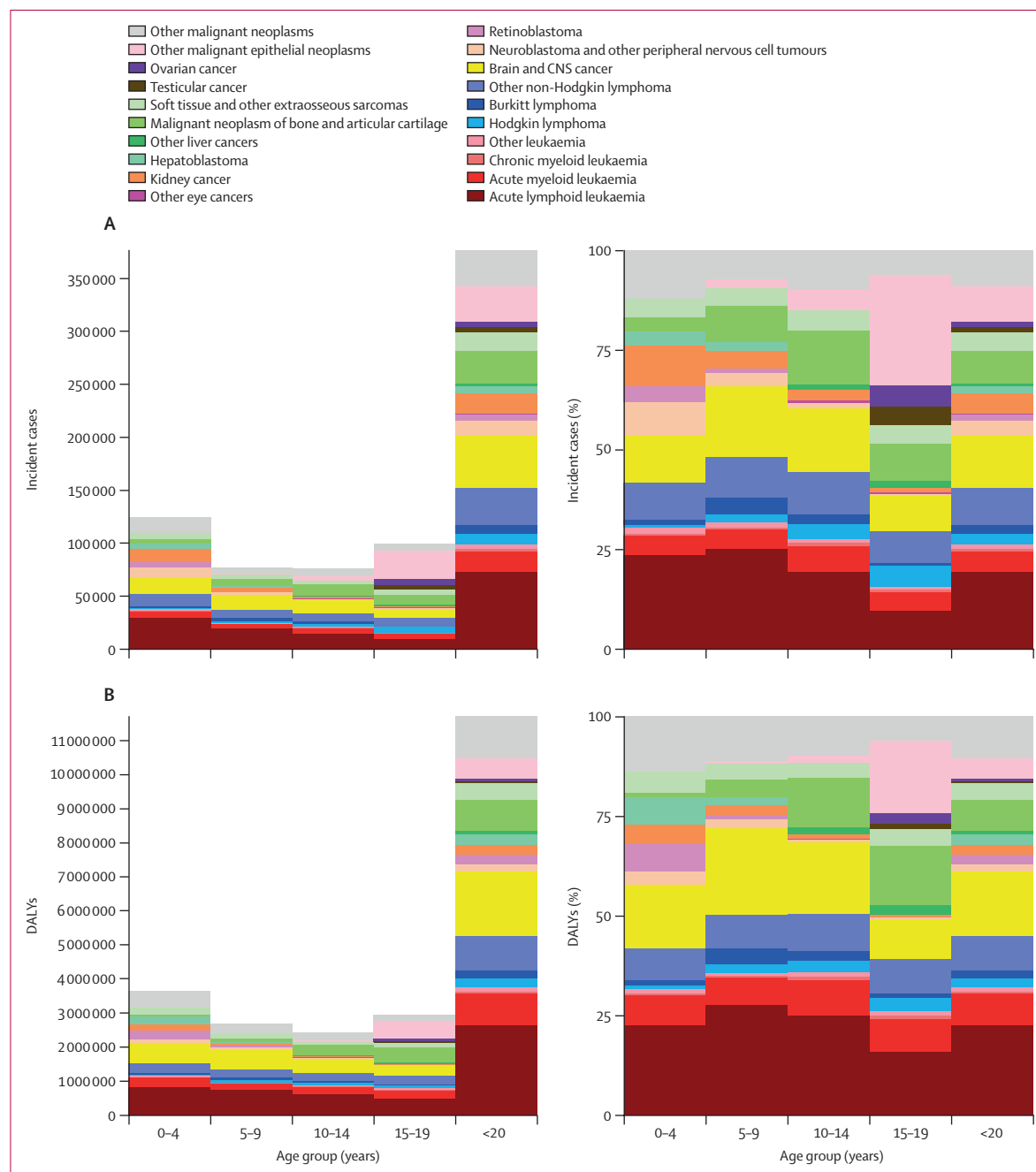


Figure 1: Global childhood cancer incidence and DALY burden in 2023, by 5-year age group and overall for age 0–19 years, for all sexes combined (A) Number of incident cases (left) and proportion of incident cases among all cases of childhood cancer (right), by cancer type. (B) Number of DALYs (left) and proportion of DALYs among all childhood cancer DALYs (right), by cancer type. The <20 years age group represents the total of the age-specific categories. DALYs=disability-adjusted life-years.

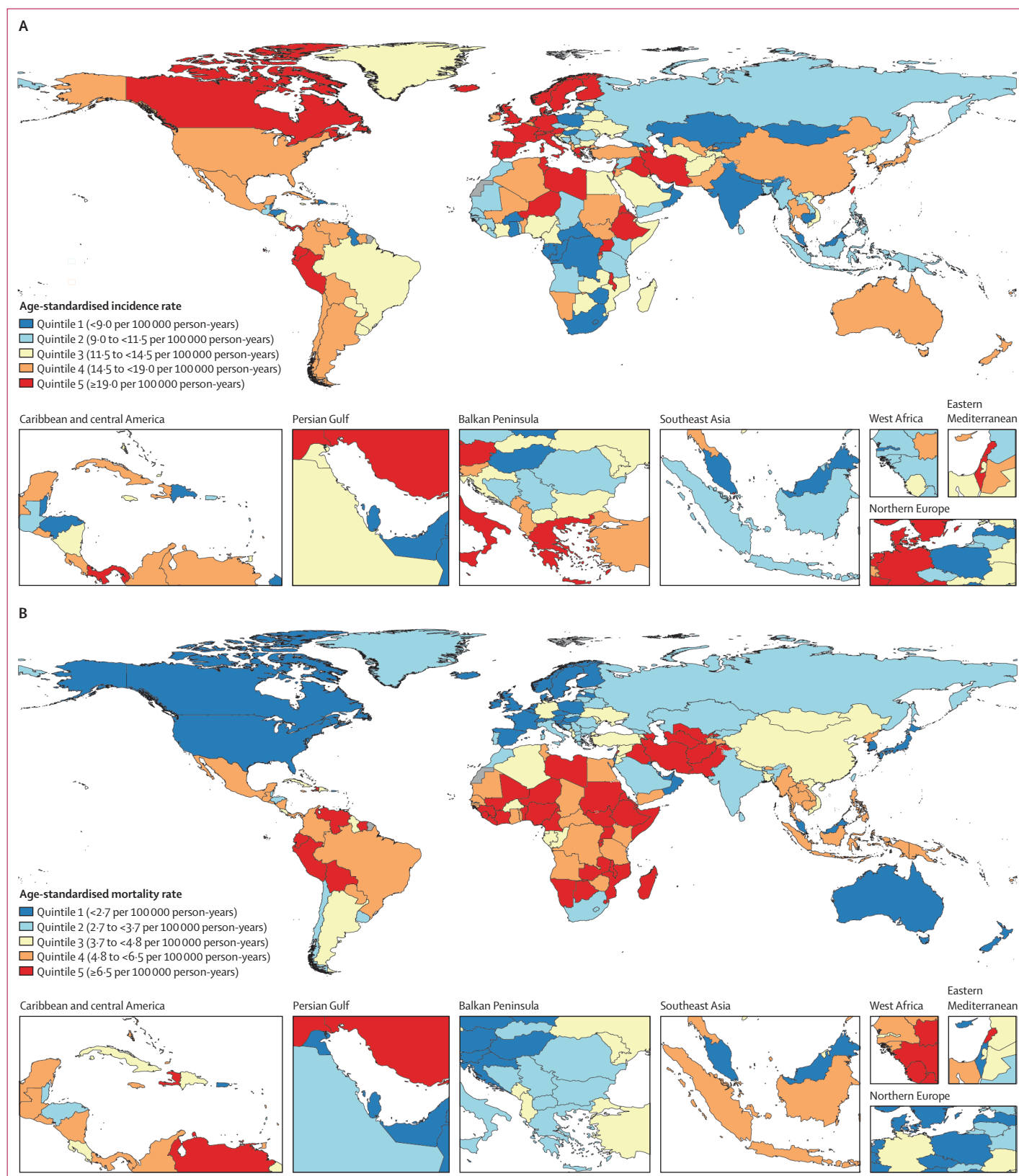


Figure 2: Global map of age-standardised childhood cancer incidence rates (A) and mortality rates (B) in 2023, for all sexes combined

notably in high and high-middle SDI settings (appendix 2 p 32).

Across the childhood cancer age groups globally in 2023, the greatest numbers of incident cases and DALYs occurred in the 0–4 years age group, with 124 000 cases (95% UI 92 200–169 000) and 3.65 million (3.20–4.22) DALYs, followed by the 15–19 years age group (99 400 cases [78 500–123 000] and 2.96 million [2.66–3.29] DALYs; figure 1). The majority of childhood cancer DALY burden for ages 0–19 years was from leukaemias (31.9% [28.2–37.1]), brain and CNS cancers (16.2% [13.6–19.7]), non-Hodgkin lymphoma (10.6%

[8.2–13.8]), and other malignant neoplasms (ie, uncategorised cancers; 10.5% [8.1–13.1]). The contribution of solid tumours to total cancer DALY burden was higher in the 0–4 years (28.2% [22.9–33.7]) and 15–19 years (44.8% [41.3–48.8]) age groups than in the 5–9 years (16.9% [14.1–19.3]) and 10–14 years (21.6% [17.6–25.9]) age groups. The proportion of uncategorised cancers (other malignant neoplasms) was greatest in the 0–4 years age group at 14.1% (10.5–17.9). The majority of global childhood cancer DALY burden in 2023 was composed of YLLs, at 98.5% (97.6–99.1; appendix 2 p 40).

The global burden of several cancer types estimated for the first time in GBD in the 2021 iteration, and presented for childhood and adolescent age groups for the first time in this GBD 2023 analysis, are detailed in appendix 2 (p 41). These new cancer causes (Burkitt lymphoma, other non-Hodgkin lymphoma, soft tissue and other extraosseous sarcomas, malignant neoplasms of bone and articular cartilage, neuroblastoma and other peripheral nervous cell tumours, hepatoblastoma, retinoblastoma, and other eye cancers) comprised 29.6% (95% UI 25.9–33.2) of childhood cancer DALYs globally in 2023 (figure 1). Among this newly modelled burden, malignant neoplasms of bone and articular cartilage accounted for a particularly large proportion of overall childhood cancer DALYs in the 10–14 years (12.3% [8.4–16.6]) and 15–19 years (14.8% [10.6–19.4]) age groups, while other non-Hodgkin lymphoma comprised the largest proportion among the new cancer types in the 0–4 years (7.9% [5.8–11.2]) and 5–9 years (8.5% [6.2–11.1]) age groups (figure 1).

The global geographical distribution of age-standardised childhood cancer incidence rates in 2023 was most prominent in HICs, while age-standardised childhood cancer mortality rates was weighted towards lower-income countries in the WHO African and Eastern Mediterranean regions (table; figure 2; appendix 2 p 15). The WHO regions with greatest mortality also tended to be regions with sparse data and consequently greater uncertainty in estimates (appendix 2 pp 2–5). In 2023, age-standardised YLLs due to childhood cancer, reflective of premature mortality, were inversely correlated with country-level SDI (Pearson correlation coefficient -0.658 [95% UI -0.708 to -0.604]). There was generally a wider spread of age-standardised YLL rates in lower SDI settings than higher SDI settings (figure 3).

Childhood cancer was the eighth-leading cause of childhood death globally in 2023, following neonatal disorders, congenital birth defects, road injuries, and several communicable diseases such as lower respiratory infections and malaria, but with greater burden than other communicable diseases such as measles, tuberculosis, and HIV/AIDS (figure 4A). The contribution of childhood cancer to childhood deaths in 2023, as represented by ranking of the number of deaths compared with those caused by other diseases and injuries, was greater in higher SDI settings,

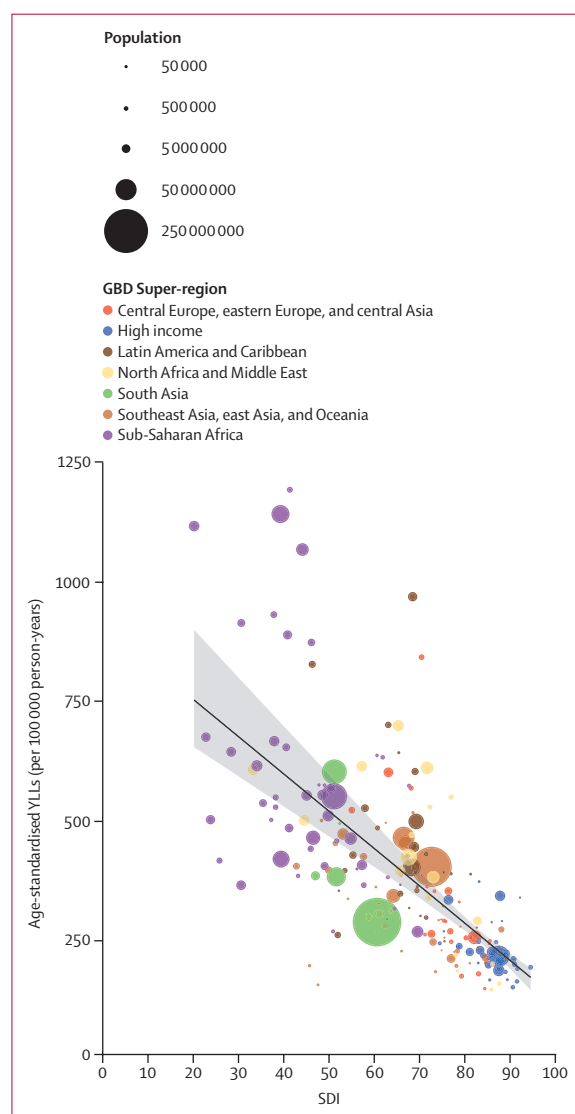


Figure 3: The association between SDI and age-standardised rates of childhood cancer YLLs in 2023, for all sexes combined

Results are presented at the country level. Each circle represents one country, with size relative to the country's total population aged 0–19 years. The grey line and shading represent the mean and 95% UI, respectively, for a linear regression between SDI and the age-standardised YLL rate (Pearson correlation coefficient -0.675 [95% UI -0.723 to -0.618]). SDI=Socio-demographic Index. UI=uncertainty interval. YLLs=years of life lost.

ranking third in high SDI and fifth in high-middle SDI quintiles, and lower in lower SDI settings, ranking seventh in the middle SDI and low-middle SDI quintiles, and 13th in the low SDI quintile (figure 4A). Childhood cancer was the 11th-leading cause of childhood DALYs globally in 2023, with this burden ranking higher in high SDI and high-middle SDI quintile settings compared with other SDI quintiles (appendix 2 p 6).

When compared with cancers occurring in adults (age ≥ 20 years), childhood cancers as a group ranked 18th in deaths globally in 2023 out of 37 cancers, but seventh in low SDI quintile settings (appendix 2 p 7). Childhood cancer burden expressed in DALYs was the ninth-leading cause of total cancer DALYs globally in 2023, following more common or more fatal cancers of adulthood such as lung cancer, colon and rectum cancer, breast cancer, and stomach cancer (figure 4B). In lower SDI quintile settings, childhood cancer DALYs were ranked higher, ranking second in low SDI and sixth in low-middle SDI and middle SDI settings.

Target cancers for the WHO GICC comprised 47.3% (95% UI 42.2–52.0) of global childhood cancer deaths in 2023 (figure 5). This percentage varied by WHO world region, with greater contributions of childhood cancer deaths in the region of the Americas of 55.7% (53.6–57.3) and lesser contributions in the South-East Asia region of 42.1% (35.9–49.4). Brain and CNS cancers and acute lymphoid leukaemia comprised the largest proportion of deaths out of the WHO index cancers in each region. The smallest percentages of childhood cancer due to these two cancers combined

were in the African region (28.6% [23.7–34.0]) and South-East Asia region (36.6% [30.7–43.8]), with the other WHO GICC indicator cancers contributing greater proportional deaths in the African region than in the other WHO world regions (figure 5). Globally, of the WHO GICC indicator cancers, acute lymphoid leukaemia contributed the most deaths (32 200 [24 300–41 100]) in 2023, followed by brain and CNS cancers (23 200 [18 900–29 000]), Hodgkin lymphoma (3310 [2090–4670]), retinoblastoma (3230 [1270–7010]), kidney cancer (3190 [2410–3960]), and Burkitt lymphoma (2960 [1960–4930]; appendix 2 p 43). Further detailed estimates for these cancer types, and all cancer types reported in this analysis, globally and by WHO world region, are available in appendix 2 (pp 44–51).

Discussion

Since the completion of the GBD 2017 analysis on global childhood cancer burden, major health policy and academic initiatives have shifted the landscape of childhood cancer control worldwide. First, WHO and multiple global partners launched the GICC in 2018.⁴ Since then, more than 75 governments have demonstrated a commitment to implementing childhood cancer control programmes either as WHO GICC focus countries or through activities aligned with the initiative's CureAll framework.²⁷ Second, with the launch of the St Jude Global Alliance²⁸ in 2018, over 400 hospitals and non-profit foundations in more than 90 countries are working collaboratively to implement novel and evidence-based interventions^{29,30} and improve health-care delivery services for childhood cancer at the institutional level.

A		Deaths (95% UIs)	Rank					
			Global	High SDI	High-middle SDI	Middle SDI	Low-middle SDI	Low SDI
	Neonatal disorders	1630 000 (1510 000–1750 000)	1	1	1	1	1	1
	Lower respiratory infections	712 000 (588 000–855 000)	2	6	3	3	2	2
	Congenital birth defects	510 000 (396 000–647 000)	3	2	2	2	3	5
	Malaria	473 000 (180 000–876 000)	4	92	17	5	5	3
	Diarrhoeal diseases	380 000 (279 000–525 000)	5	19	6	6	4	4
	Road injuries	220 000 (170 000–282 000)	6	4	4	4	6	7
	Meningitis	148 000 (112 000–208 000)	7	21	18	10	8	8
	Childhood cancer	144 000 (131 000–162 000)	8	3	5	7	7	13
	Measles	142 000 (57 700–252 000)	9	52	36	26	20	6
	Drowning	127 000 (95 300–166 000)	10	9	7	8	9	11
	Pertussis	113 000 (65 000–185 000)	11	32	14	12	12	10
	Protein-energy malnutrition	105 000 (75 500–142 000)	12	45	13	24	16	9
	Tuberculosis	87 800 (59 400–124 000)	13	47	22	16	14	12
	Sexually transmitted infections excluding HIV	74 100 (27 100–143 000)	14	35	16	15	17	15
	HIV/AIDS	72 000 (61 900–84 700)	15	38	11	29	21	14
	Interpersonal violence	68 900 (57 200–87 000)	16	10	8	13	15	19
	Self-harm	63 400 (53 500–76 300)	17	7	9	20	11	26
	Typhoid and paratyphoid	60 500 (31 900–101 000)	18	48	20	23	10	17
	Invasive non-typhoidal salmonella	59 300 (43 000–80 500)	19	77	53	44	28	16
	COVID-19	52 900 (48 300–57 800)	20	15	15	9	30	23

(Figure 4 continues on next page)

B

	DALYs (95% UIs)	Rank					
		Global	High SDI	High-middle SDI	Middle SDI	Low-middle SDI	Low SDI
Tracheal, bronchus, and lung cancer	46 600 000 (42 400 000–50 700 000)	1	1	1	1	2	7
Colon and rectum cancer	26 100 000 (23 700 000–28 500 000)	2	2	3	3	7	6
Breast cancer	24 500 000 (21 600 000–28 000 000)	3	4	4	2	1	3
Stomach cancer	22 500 000 (19 200 000–25 900 000)	4	3	2	4	5	4
Oesophageal cancer	14 000 000 (12 600 000–15 800 000)	5	6	5	10	4	8
Liver cancer	13 500 000 (11 800 000–15 500 000)	6	7	6	5	9	5
Cervical cancer	13 100 000 (10 000 000–17 100 000)	7	15	7	7	3	1
Pancreatic cancer	12 300 000 (11 400 000–13 100 000)	8	5	8	8	17	16
Childhood cancer	11 700 000 (10 700 000–13 200 000)	9	20	9	6	6	2
Prostate cancer	8 910 000 (7 870 000–10 000 000)	10	8	12	12	11	10
Brain and CNS cancer	7 260 000 (6 340 000–8 670 000)	11	9	10	9	16	17
Non-Hodgkin lymphoma	7 080 000 (6 160 000–8 080 000)	12	10	13	13	13	11
Lip and oral cavity cancer	6 420 000 (5 540 000–7 570 000)	13	18	11	15	8	9
Ovarian cancer	6 400 000 (5 400 000–7 540 000)	14	12	14	11	10	12
Other malignant neoplasms	5 460 000 (4 720 000–6 110 000)	15	14	15	14	14	13
Bladder cancer	4 620 000 (4 220 000–5 120 000)	16	11	18	18	21	20
Gallbladder and biliary tract cancer	4 000 000 (3 460 000–4 800 000)	17	16	16	21	18	18
Kidney cancer	3 780 000 (3 390 000–4 200 000)	18	13	20	24	28	30
Larynx cancer	3 560 000 (2 990 000–4 290 000)	19	23	17	19	15	15
Other pharynx cancer	3 330 000 (2 680 000–4 160 000)	20	24	19	23	12	14
Acute myeloid leukaemia	3 310 000 (2 800 000–3 920 000)	21	17	21	17	22	23
Uterine cancer	2 900 000 (2 470 000–3 390 000)	22	21	25	22	23	26
Multiple myeloma	2 810 000 (2 540 000–3 130 000)	23	19	24	28	24	28
Nasopharynx cancer	2 500 000 (2 080 000–2 950 000)	24	25	22	16	19	19
Malignant neoplasm of bone and articular cartilage	2 120 000 (1 680 000–2 690 000)	25	28	23	20	20	21
Malignant skin melanoma	1 810 000 (1 600 000–2 130 000)	26	22	30	32	32	31
Acute lymphoid leukaemia	1 740 000 (1 150 000–2 350 000)	27	30	26	25	29	25
Soft tissue and other extrasosseous sarcomas	1 700 000 (1 380 000–2 150 000)	28	26	29	29	25	22
Other leukaemia	1 690 000 (1 300 000–2 190 000)	29	29	27	26	26	29
Thyroid cancer	1 500 000 (1 260 000–1 780 000)	30	31	28	27	27	24
Chronic lymphoid leukaemia	983 000 (866 000–1 180 000)	31	27	31	34	34	35
Hodgkin lymphoma	922 000 (700 000–1 200 000)	32	34	32	31	30	27
Chronic myeloid leukaemia	720 000 (562 000–922 000)	33	33	34	30	31	32
Mesothelioma	624 000 (553 000–695 000)	34	32	35	35	35	36
Testicular cancer	552 000 (443 000–686 000)	35	35	33	33	33	33
Eye cancer	196 000 (150 000–261 000)	36	36	37	36	36	34
Neuroblastoma and other peripheral nervous cell tumours	121 000 (104 000–147 000)	37	37	36	37	37	37

Figure 4: Contribution of childhood cancer to global child deaths and all-age cancer DALYs in 2023, for all sexes combined

(A) Rankings by cause comparing the absolute number of global deaths among the top 20 Level 3 categories of disease in GBD in ages 0–19 years. (B) Rankings by cause comparing the absolute number of global DALYs in GBD for childhood cancer (all cancers for ages 0–19 years) with individual cancer causes for ages 20 years to ≥95 years (these cancer-specific rows exclude DALYs for ages 0–19 years). For both sets of rankings, causes are ordered in descending absolute value of the global estimates, with childhood cancer in bold. Cells are coloured according to rank, ranging from highest rank (red) to lowest rank (green). The final five columns provide the rankings of these top global causes within each SDI quintile. DALYs=disability-adjusted life-years. GBD=Global Burden of Diseases, Injuries, and Risk Factors Study. SDI=Socio-demographic Index. UI=uncertainty interval.

Third, efforts to expand capacity for local childhood cancer data collection and dissemination were initiated in 2020 with the childhood offshoot of the International Agency for Research on Cancer's Global Initiative for Cancer Registries (ChildGICR),^{15,16} which holds great promise for improving the currently inadequate observed cancer surveillance data globally.¹³ More recently, the 2020 *Lancet Oncology* Commission on sustainable care for children with cancer³¹ was completed, providing an investment case and roadmap to monitoring childhood cancers through a proposed indicator framework focused on reducing financial hardship for patients and families, increasing access to multimodal therapy, and ensuring quality in the provision of health services. Underlying

this framework and the WHO initiative, however, are the need for improved epidemiological data to better inform stakeholders interested in improving childhood cancer outcomes through health systems and health service delivery interventions. Estimates of international and national cancer burden patterns are important to inform these efforts as well as national cancer control plans, but not every country has the surveillance systems necessary to generate these data.

To contribute to the broader ongoing efforts to improve outcomes for childhood cancers globally, we updated estimates of global, regional, and national childhood cancer burden across time and incorporated cancer types with substantial burden in childhood and adolescent

years, such as Burkitt lymphoma, soft tissue sarcomas, malignant bone tumours, retinoblastoma, hepatoblastoma, and neuroblastoma. This is the first time that a global estimation group has provided a formal analysis of estimates for these key cancers relevant to WHO GICC goals, building to the potential to monitor global childhood cancer control efforts and providing information relevant to childhood cancer policy development. There were an estimated 144 000 deaths (95% UI 131 000–162 000) due to childhood cancer in 2023, and target cancers for the WHO GICC comprised approximately 47·3% (42·2–52·0) of global childhood cancer deaths. With the addition of the nine new cancer types into the GBD framework, only 10·5% (8·1–13·1) of childhood cancer DALYs remain uncategorised, compared with approximately 26·5% (26·5–26·5) in GBD 2017. Childhood cancers represent a spectrum of diseases, with each requiring diverse health services ranging from chemotherapy and radiation access to different surgical subspecialty services, such as paediatric surgery, ophthalmology, orthopaedics, or neurosurgery. With greater granularity of estimates by cancer types, health-care providers and policy makers can plan human resource requirements, referral networks, and financial products for cancer subtypes included in the GICC, such as retinoblastoma and Burkitt lymphoma.

The GBD framework also provides estimates of DALYs for each cancer type and childhood cancers in total, with 11·7 million (95% UI 10·7–13·2) DALYs estimated globally for childhood cancers in 2023. The *Lancet Oncology* Commission on sustainable care for children with cancer³¹ advocated for the inclusion of childhood cancers in essential benefits packages and financial risk protection for families of children with cancer through the elimination of out-of-pocket expenditures. To meet this goal, health policy makers and economists often rely on GBD-estimated DALYs, in combination with other data, to determine which interventions to cover as part of their universal health coverage scale-up.^{32–37} Decision making regarding health benefits packages and broader health policy priorities are complex and context dependent, but data on the burden of childhood cancer relative to other diseases of childhood and to adult cancers might be useful for some countries. We found that childhood cancers were the eighth-leading cause of childhood death and the ninth-leading cause of total cancer DALYs globally in 2023, with variation by SDI. These comprehensive and more granular new results will be important for the ongoing implementation work associated with the WHO GICC and for the many global partners working on childhood cancer control initiatives locally and internationally.

From a broader perspective, GBD 2023 estimates continue to show persistent inequities in childhood cancer burden globally. In 2023, age-standardised YLL rates due to childhood cancers were inversely correlated

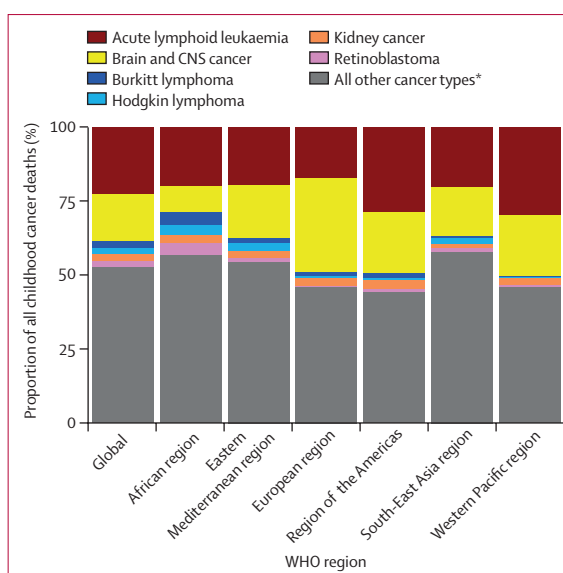


Figure 5: Proportion of total childhood cancer deaths due to index cancers for the WHO GICC in 2023, globally and by WHO region, for all sexes combined

Childhood cancer is defined as that occurring at ages 0–19 years. GICC=Global Initiative for Childhood Cancer. *All other cancer types includes all those not reported separately as WHO GICC index cancers.

with country-level SDI, indicating that premature mortality due to childhood cancer is greatest in countries with the least resources available to address it. This finding is consistent with an analysis of childhood cancer burden using GBD 2017 estimates, which found that age-standardised mortality rates were weighted towards settings with lower SDI, and underscores the need for global collaboration and investment in childhood cancer that targets support for LMICs. Additionally, deaths due to childhood cancers decreased by 27·0% (15·5–36·1) globally between 1990 and 2023. During the same time period, overall childhood deaths decreased by 56·5% (55·8 to 57·3) and childhood cancer incident cases changed by –0·9% (–27·1 to 38·6). While these findings suggest progress in childhood cancer care globally for children effectively diagnosed, this progress has been unbalanced, with childhood cancer deaths increasing in the African region by 55·6% (25·5–92·4), from 31 500 (24 900–38 500) in 1990 to 49 000 (42 600–58 200) in 2023. This increase in childhood cancer deaths is likely partially driven by regional population changes, as childhood cancer age-standardised mortality rates decreased globally and in all WHO world regions. Additionally, although global efforts to address childhood cancer burden are ongoing, the distal impacts of current global childhood cancer control efforts will take years to visualise in GBD models. Further analyses forecasting the burden of childhood cancer over the coming decades will be important, allowing for national and international policy makers to project the changing disease burden and adjust policies and interventions accordingly.

While the estimates of the burden of childhood cancers produced by GBD 2023 are valuable contributions to global childhood cancer control, there are limitations as well as future opportunities to expand estimation efforts. First, while UIs are estimated and provided with all measures in GBD, there is greater uncertainty in countries that generally have fewer data available and likely fewer resources to support the recording and reporting of these data. Additionally, UI estimates cannot account for every potential bias. In GBD, corrections to the data are made to address measurement error and, in cause-of-death estimation, ensemble modelling is used to deal with model misspecification bias as much as possible; however, residual bias affecting results cannot be excluded. Relatedly, because models are fitted across each cancer cause's entire age range, data for age groups older than 19 years might influence childhood cancer estimates, particularly when data are unavailable in childhood age groups. These challenges underscore the importance of reliable incidence and mortality data to inform childhood cancer burden estimates, and efforts under way through the ChildGICR initiative¹⁵ to establish and improve childhood cancer registration systems are crucial to inform our understanding of the burden of childhood cancer. As the ability to capture and report more detailed aspects of childhood cancer diagnoses and outcomes develops, incorporating these data into future GBD rounds would be expected to improve estimates of childhood cancer burden locally, regionally, and globally.

A second limitation of this study is that, given delays in population-based data availability during the COVID-19 pandemic, this analysis did not incorporate changes in childhood cancer burden due to the influence of the pandemic. Reports thus far suggest a decrease in diagnosis of childhood cancers in some locations and reduced access to treatment services during the early pandemic period, with treatment access challenges greater in LMICs.^{38,39} As cancer surveillance data continue to emerge with adequate diversity in geographical representation and across the span of time, understanding the effect of the COVID-19 pandemic on childhood cancer burden on a global scale will be important to informing future preparation endeavours. Third, GBD estimates currently do not separately report burden due to low-grade gliomas, a childhood cancer subgroup that causes substantial burden in children⁴⁰ and a GICC index cancer type. Our current analysis used malignant CNS cancer burden as a surrogate for low-grade gliomas and thus likely overestimated the fatal burden of WHO target cancers in total. Similarly, several new GBD causes are based on ICDC-3 diagnostic groups, such as malignant bone tumours and soft tissue sarcomas, and are not currently disaggregated into clinically important subgroups such as rhabdomyosarcoma, osteosarcoma, or Ewing sarcoma

and related sarcomas of the bone. Fourth, as access to and quality of childhood cancer services improves worldwide, there will be an increasing number of survivors, and the current GBD framework does not estimate the lifelong disease burden anticipated in this growing population. Efforts to address this challenging limitation are under consideration. Fifth, current GBD estimation cannot completely account for cancer mimics that might result in misdiagnosis in children and potentially inaccurate cause-of-death data.¹³ Finally, GBD does not currently externally report estimates of childhood cancer survival, and previous modelling efforts have been single publications that are not updated over time with new data or improved methods. The generation of global childhood cancer survival estimates in a continuously updated modelling framework that routinely produces estimates across time, such as GBD, would allow for formal tracking of WHO GICC progress where survival data are not currently available, and should be a priority in the future. This does not negate the importance of supporting and generating childhood cancer survival results as locally as possible around the world, but would aid country planning where this information is not available, and would help overall global tracking efforts. More general limitations of the GBD study have been discussed in other publications.^{14,18,19,21}

In summary, this updated analysis of the global burden of childhood cancer provides estimates for 20 cancer categories—including for nine additional cancer causes compared with previous GBD analyses, which represent important cancers in children—from 1990 to 2023. The addition of new cancer types relevant to children is an important step towards aligning with WHO GICC targets. This study highlights that childhood cancer persists as a substantial contributor to health loss in children, and that considerable disparities in childhood cancer burden persist globally across a range of demographic factors, including by SDI and world region. We found that resource-limited settings are disproportionately burdened by childhood cancer, indicating that future efforts to decrease the global childhood cancer burden must focus on addressing inequities and expanding access to comprehensive cancer services. We anticipate that these estimates will enhance efforts by UN agencies, governments, and other relevant childhood cancer stakeholders to design and implement cancer control programmes and interventions worldwide.

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See appendix 3 (pp 3–10) for collaborators' affiliations.

See Online for appendix 3

Contributors

See appendix 3 (pp 10–13) for more detailed information about individual author contributions to the research, divided into the following categories: managing the overall research enterprise; writing the first draft of the manuscript; primary responsibility for applying analytical methods to produce estimates; primary responsibility for seeking, cataloguing, extracting, or cleaning data; designing or coding figures and tables; providing data or critical feedback on data sources; developing methods or computational machinery; providing critical feedback on methods or results; drafting the manuscript or revising it critically for important intellectual content; and managing the estimation or publications process. The corresponding authors (L M Force and N Bhakta) had final responsibility for the decision to submit for publication. Members of the Institute for Health Metrics and Evaluation Cancer Team had full access to the underlying data used to generate estimates presented in this Article. All other authors had access to and reviewed estimates as part of the research evaluation process, which includes additional stages of formal review.

Declaration of interests

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Data sharing

To download the data used in these analyses, please visit the Global Health Data Exchange (<https://ghdx.healthdata.org/gbd-2023>). The

statistical code used for data processing and analysis is available upon request; requests should be sent to engage@healthdata.org.

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References

- 1 Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. *CA Cancer J Clin* 2023; **73**: 17–48.
- 2 Botta L, Gatta G, Capocaccia R, et al, and the EURO CARE-6 Working Group. Long-term survival and cure fraction estimates for childhood cancer in Europe (EURO CARE-6): results from a population-based study. *Lancet Oncol* 2022; **23**: 1525–36.
- 3 Ward ZJ, Yeh JM, Bhakta N, Frazier AL, Girardi F, Atun R. Global childhood cancer survival estimates and priority-setting: a simulation-based analysis. *Lancet Oncol* 2019; **20**: 972–83.
- 4 WHO. The Global Initiative for Childhood Cancer. <https://www.who.int/initiatives/the-global-initiative-for-childhood-cancer> (accessed Oct 17, 2023).
- 5 Union for International Cancer Control. Global Initiative for Cancer Registries (GICR). <https://www.uicc.org/what-we-do/sharing-knowledge/global-initiative-cancer-registries-gicr> (accessed Jan 8, 2024).
- 6 Siddiqui AH, Zafar SN. Global availability of cancer registry data. *J Glob Oncol* 2018; **4**: 1–3.
- 7 Steliarova-Foucher E, Colombet M, Ries LAG, et al, and the IICC-3 contributors. International incidence of childhood cancer, 2001–10: a population-based registry study. *Lancet Oncol* 2017; **18**: 719–31.
- 8 Huang J, Chan SC, Ngai CH, et al, and the NCD Global Health Research Group, Association of Pacific Rim Universities (APRU). Global incidence, mortality and temporal trends of cancer in children: a joinpoint regression analysis. *Cancer Med* 2023; **12**: 1903–11.
- 9 Allemani C, Matsuda T, Di Carlo V, et al, and the CONCORD Working Group. Global surveillance of trends in cancer survival 2000–14 (CONCORD-3): analysis of individual records for 37 513 025 patients diagnosed with one of 18 cancers from 322 population-based registries in 71 countries. *Lancet* 2018; **391**: 1023–75.
- 10 Johnston WT, Erdmann F, Newton R, Steliarova-Foucher E, Schütz J, Roman E. Childhood cancer: estimating regional and global incidence. *Cancer Epidemiol* 2021; **71**: 101662.
- 11 GBD 2017 Childhood Cancer Collaborators. The global burden of childhood and adolescent cancer in 2017: an analysis of the Global Burden of Disease Study 2017. *Lancet Oncol* 2019; **20**: 1211–25.
- 12 Sharma R. A systematic examination of burden of childhood cancers in 183 countries: estimates from GLOBOCAN 2018. *Eur J Cancer Care (Engl)* 2021; **30**: e13438.
- 13 Bhakta N, Force LM, Allemani C, et al. Childhood cancer burden: a review of global estimates. *Lancet Oncol* 2019; **20**: e42–53.
- 14 GBD 2023 Causes of Death Collaborators. Global burden of 292 causes of death in 204 countries and territories and 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet* 2025; **406**: 1811–72.
- 15 WHO, International Agency for Research on Cancer. Targeting childhood cancer through the Global Initiative for Cancer Registry Development (ChildGICR). <https://gicr.iarc.fr/child-gicr/> (accessed Jan 8, 2024).
- 16 WHO, International Agency for Research on Cancer. The Global Initiative for Cancer Registry Development. <https://gicr.iarc.fr/about-the-gicr/> (accessed Oct 17, 2023).
- 17 Institute for Health Metrics and Evaluation. Protocol for the Global Burden of Diseases, Injuries, and Risk Factors Study (GBD). <https://www.healthdata.org/research-analysis/about-gbd/protocol> (accessed July 9, 2024).
- 18 GBD 2023 Disease and Injury and Risk Factor Collaborators. Burden of 375 diseases and injuries, risk-attributable burden of 88 risk factors, and healthy life expectancy in 204 countries and territories, including 660 subnational locations, 1990–2023: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet* 2025; **406**: 1873–922.
- 19 GBD 2023 Demographics Collaborators. Global age-sex-specific all-cause mortality and life expectancy estimates for 204 countries and territories and 660 subnational locations, 1950–2023: a demographic analysis for the Global Burden of Disease Study 2023. *Lancet* 2025; **406**: 1731–810.
- 20 Stevens GA, Alkema L, Black RE, et al, and The GATHER Working Group. Guidelines for Accurate and Transparent Health Estimates Reporting: the GATHER statement. *Lancet* 2016; **388**: e19–23.
- 21 Force LM, Kocarnik JM, May ML, et al, and the GBD 2023 Cancer Collaborators. The global, regional, and national burden of cancer, 1990–2023, with forecasts to 2050: a systematic analysis for the Global Burden of Disease Study 2023. *Lancet* 2025; **406**: 1565–86.
- 22 Foreman KJ, Lozano R, Lopez AD, Murray CJ. Modeling causes of death: an integrated approach using CODEm. *Popul Health Metr* 2012; **10**: 1.
- 23 WHO. ICD-10: international statistical classification of diseases and related health problems: tenth revision, 2nd edn. 2004. <https://iris.who.int/handle/10665/42980> (accessed July 18, 2024).
- 24 Steliarova-Foucher E, Stiller C, Lacour B, Kaatsch P. International Classification of Childhood Cancer, third edition. *Cancer* 2005; **103**: 1457–67.
- 25 WHO. Countries. <https://www.who.int/countries> (accessed Oct 17, 2023).
- 26 The World Bank. World Bank country and lending groups. <https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups> (accessed July 1, 2024).
- 27 WHO. CureAll framework: WHO global initiative for childhood cancer: increasing access, advancing quality, saving lives. World Health Organization, 2021. <https://iris.who.int/handle/10665/347370> (accessed July 9, 2024).
- 28 St. Jude Children's Research Hospital. About St. Jude Global Alliance. <https://global.stjude.org/en-us/global-alliance/about.html> (accessed July 18, 2024).
- 29 Agulnik A, Muniz-Talavera H, Pham LTD, et al, and the EVAT Study Group. Effect of paediatric early warning systems (PEWS) implementation on clinical deterioration event mortality among children with cancer in resource-limited hospitals in Latin America: a prospective, multicentre cohort study. *Lancet Oncol* 2023; **24**: 978–88.
- 30 McNeil MJ, Garcia Quintero X, Gonzalez M, et al. Preventing and treating pain and anxiety during needle-based procedures in children with cancer in low- and middle-income countries. *Cancers (Basel)* 2024; **16**: 1025.
- 31 Atun R, Bhakta N, Denburg A, et al. Sustainable care for children with cancer: a Lancet Oncology Commission. *Lancet Oncol* 2020; **21**: e185–224.
- 32 Institute for Health Metrics and Evaluation. Government of Ethiopia using Global Burden of Disease evidence to achieve equity and access to health care for all. <https://www.healthdata.org/news-events/insights-blog/acting-data/government-ethiopia-using-global-burden-disease-evidence> (accessed Oct 13, 2023).
- 33 Institute for Health Metrics and Evaluation. Burden of disease data help to illuminate new policy priorities in Rwanda. <https://www.healthdata.org/news-events/insights-blog/acting-data/burden-disease-data-help-illuminate-new-policy-priorities> (accessed Oct 16, 2023).
- 34 Institute for Health Metrics and Evaluation. Using burden of disease data to prioritize national health resources in the Philippines. <https://www.healthdata.org/news-events/insights-blog/acting-data/using-burden-disease-data-prioritize-national-health> (accessed Oct 16, 2023).
- 35 Institute for Health Metrics and Evaluation. Understanding the 'big picture' can lead to a healthier population in Ghana. <https://www.healthdata.org/news-events/insights-blog/acting-data/understanding-big-picture-can-lead-healthier-population-ghana> (accessed Oct 16, 2023).
- 36 Institute for Health Metrics and Evaluation. Disease burden data guide spending, health policy in Mexico. <https://www.healthdata.org/news-events/insights-blog/acting-data/disease-burden-data-guide-spending-health-policy-mexico> (accessed Oct 16, 2023).

-
- 37 Institute for Health Metrics and Evaluation. NIH to use burden of disease data to maximize lifesaving impact of its funding. <https://www.healthdata.org/news-events/insights-blog/acting-data/nih-use-burden-disease-data-maximize-lifesaving-impact-its> (accessed Oct 16, 2023).
- 38 Graetz D, Agulnik A, Ranadive R, et al. Global effect of the COVID-19 pandemic on paediatric cancer care: a cross-sectional study. *Lancet Child Adolesc Health* 2021; **5**: 332–40.
- 39 Vasquez L, Sampor C, Villanueva G, et al. Early impact of the COVID-19 pandemic on paediatric cancer care in Latin America. *Lancet Oncol* 2020; **21**: 753–55.
- 40 Armstrong GT, Conklin HM, Huang S, et al. Survival and long-term health and cognitive outcomes after low-grade glioma. *Neuro Oncol* 2011; **13**: 223–34.